

main.py

```
1  from fastapi import FastAPI, WebSocket, WebSocketDisconnect
2  from dataclasses import dataclass
3  from typing import Optional
4
5  @dataclass
6  class Client:
7      id: int
8      name: Optional[str]
9      websocket: WebSocket
10
11  class ClientManager:
12      def __init__(self):
13          self.clients: list[Client] = []
14          self.nextId = 1
15
16      def removeConnection(self, client: Client):
17          self.clients.remove(client)
18
19      def addConnection(self, websocket: WebSocket) -> Client:
20          client = Client(
21              id = self.nextId,
22              name = None,
23              websocket = websocket
24          )
25
26          self.clients.append(client)
27          self.nextId += 1
28
29          return client
30
31      async def listen(self, client: Client):
32          try:
33              while True:
34                  data = await client.websocket.receive_json()
35                  await self.broadcast(data, client)
36
37          except WebSocketDisconnect:
38              self.removeConnection(client)
39
40      async def broadcast(self, data: str, exclusion: Client):
41          for client in self.clients:
42              if client != exclusion:
43                  await client.websocket.send_json(data)
44
45  app = FastAPI()
46  clientManager = ClientManager()
47
```

```
48 @app.websocket("/")
49 async def websocketEndpoint(websocket: WebSocket):
50     await websocket.accept()
51     client = clientManager.addConnection(websocket)
52
53     await clientManager.listen(client)
54
55 @app.get("/")
56 def get_clients():
57     return {
58         "clients": [
59             {
60                 "id": client.id,
61                 "name": client.name
62             }
63             for client in clientManager.clients
64         ]
65     }
```